

Faculty of Agriculture Chiang Mai University

ภาควิชาเกษตรศาสตร์ มหาวิทยาลัยเชียงใหม่

Faculty of Agriculture · คณะเกษตรศาสตร์



Research Team · ทีมวิจัย



Dr. Toungporn (Uttarotai)

ดร.ตวงพร (อุตรโรตไทย)

PhD in Microbiology / Highland Agriculture
Dept of Highland Agriculture & Natural
Resources

Research: AI applications in plant disease
identification

Focus: Knowledge base design · Image analysis ·
Field validation

ดร.ตวงพร (อุตรโรตไทย)

PhD in Agricultural Science

Specialisation: Crop Protection & Plant
Pathology

Research: AI applications in agricultural
extension

Focus: Knowledge base design · Guardrail policy ·
Field validation



Dr. Daranrat (Jaitiang)

ดร.ดาร์รัตน์ (ใจเที่ยง)

PhD in Agricultural Extension
Dept of Agricultural Economy & Development
Research: Digital tools for smallholder farmers

Focus: Farmer UX · Content curation · PDPA
compliance

ดร.ดาร์รัตน์ (ใจเที่ยง)

PhD in Agricultural Science

Specialisation: Soil Science & Sustainable
Agriculture

Research: Digital tools for smallholder farmers

Focus: Farmer UX · Content curation · PDPA
compliance

Project Technology Stack · เทคโนโลยีที่ใช้



Claude Sonnet 4.5

Anthropic AI · Thai language ·
Vision



RAG Pipeline

OpenAI Embeddings · Cosine
Retrieval



LINE Messaging

47M+ Thai users · Rich Menu



Guardrails

Hard safety gates · Zero
hallucination



Render Cloud

Singapore · PostgreSQL · Auto-
deploy



PDPA Compliant

Pseudonymous · Audit log · Erasure

Project Links & Contact · ข้อมูลโครงการ



Live demo: thai-farmer-agent.onrender.com



LINE bot: @643txfqa



Source code: github.com/AshyBoy91/eye-line-app



Institution: Chiang Mai University, Thailand

Development Timeline · ลำดับการพัฒนา

2026 Q1

Problem scoping: misinformation analysis in Thai farmer LINE groups; system design

2026 Q2

Core RAG pipeline + guardrail framework; knowledge base seeding (rice, soil, pests)

2026 Q2

Claude Sonnet 4.5 integration; OpenAI embeddings; LINE webhook; Render deployment

2026 Q3 (current)

Image/vision support; conversation history; daily briefing; admin LLM comparison tool; conference presentation

2026 Q4 (planned)

DOA content partnership; pgvector index; human-in-the-loop escalation; national pilot

Potential Collaboration · ความร่วมมือที่ต้องการ



Authoritative content (DOA / Rice Dept)



Provincial pilot deployment



Retrieval quality research



Farmer UX studies



Multilingual / dialect support



Research grant partnership